Network and System Management

Level 5: Step 1 - SMEs

Small and Medium Enterprises Infrastructure

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Question 1: File and Account Server Implementation

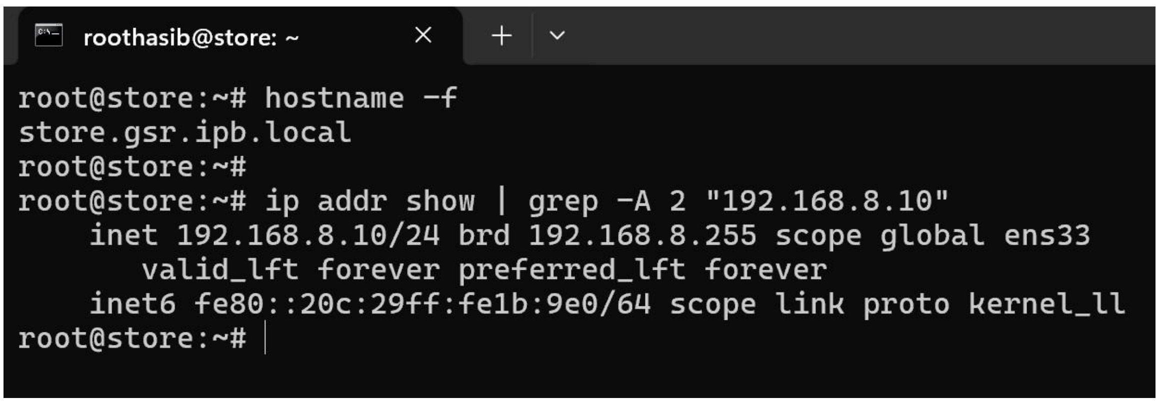
This assignment involved configuring a complete enterprise infrastructure for a Small and Medium Enterprise (SME) consisting of three virtual machines: a storage server with NFS and SNMP monitoring, a Samba Active Directory domain controller, and a Windows 10 workstation joined to the domain.

1(a) Storage Server - store.gsr.ipb.local

The storage server was implemented on a Debian 13 system providing centralized file storage, network monitoring, and SNMP services for the organization.

Hostname and Network Configuration

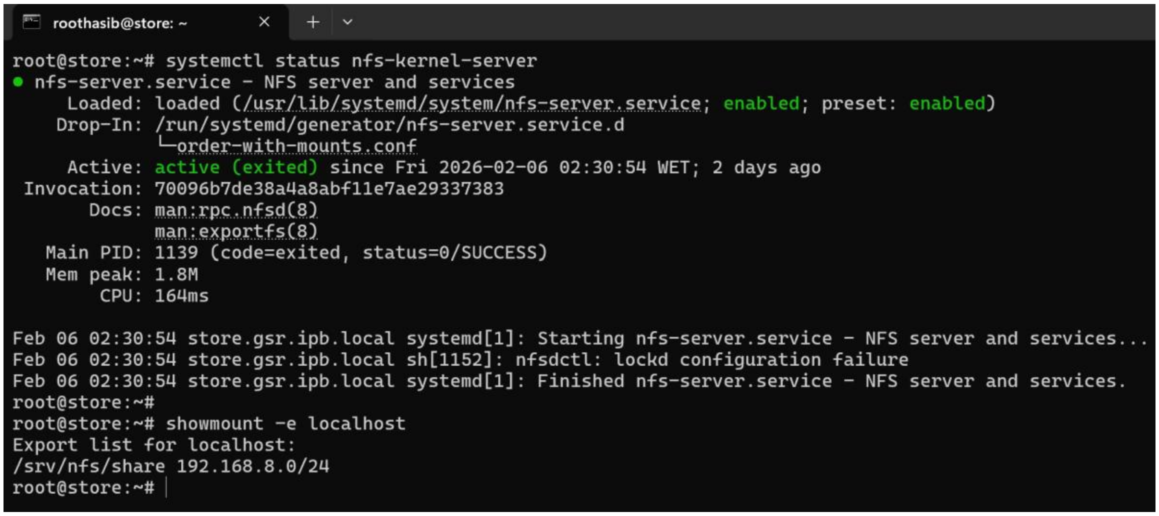
The server was configured with hostname store.gsr.ipb.local and static IP address 192.168.8.10/24 on the local network.



[INSERT SCREENSHOT 1 [1(a)]: Terminal showing hostname -f and ip addr output]

NFS Server Configuration

An NFS (Network File System) version 4 server was configured to provide shared storage accessible by other systems on the network. The NFS service exports the directory /srv/nfs/share to the entire 192.168.8.0/24 subnet, allowing the domain controller to mount this storage for backup purposes.

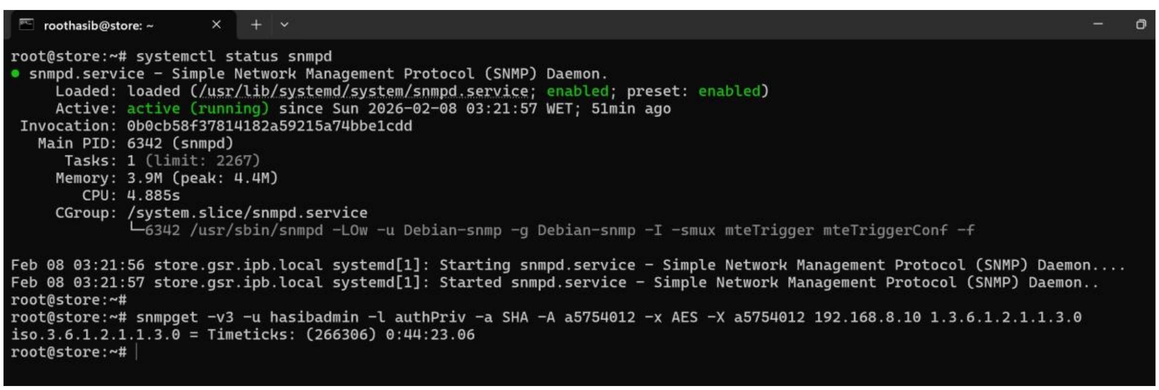


[INSERT SCREENSHOT 2 [1(a)]: NFS service status and showmount -e output]

SNMP Monitoring Service

Net-SNMP was configured with SNMPv3 authentication for secure monitoring. The SNMP service provides system metrics that can be queried by monitoring tools. SNMPv3 users were created with SHA authentication and AES encryption for enhanced security.

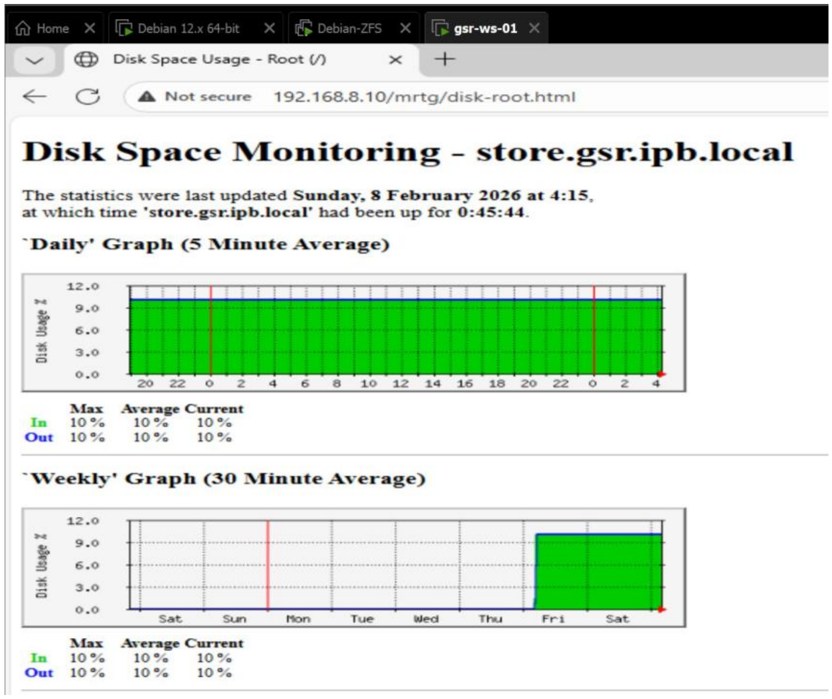
The SNMP service was verified to be responding correctly to authenticated queries, confirming proper configuration of the monitoring infrastructure.

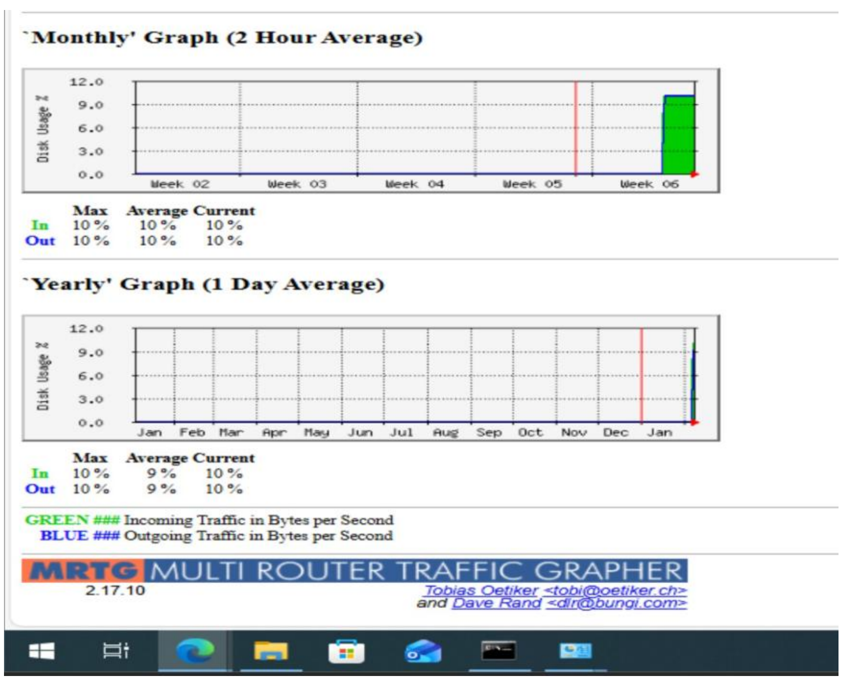


[INSERT SCREENSHOT 3 [1(a)]: SNMP service status showing active state & query verification using snmpget command]

MRTG Disk Space Monitoring

MRTG (Multi Router Traffic Grapher) was configured to monitor disk space utilization on the storage server using SNMP. The monitoring system collects data every 5 minutes via a cron job and generates graphical representations of disk usage trends accessible through a web interface.





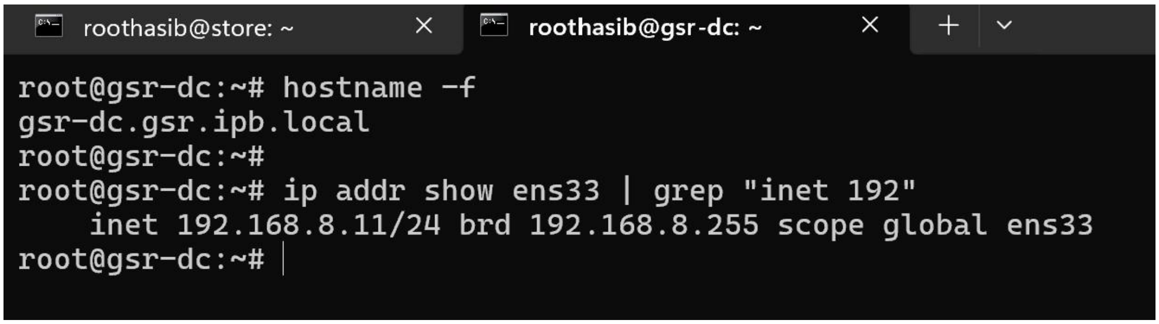
[INSERT SCREENSHOT 4 [1(a)]: MRTG web interface showing disk space monitoring graphs]

1(b) Domain Controller - gsr-dc.gsr.ipb.local

The domain controller was implemented on a Debian 13 system running Samba Active Directory Domain Controller services, providing centralized authentication, DNS services, and file sharing for the organization.

Hostname and Network Configuration

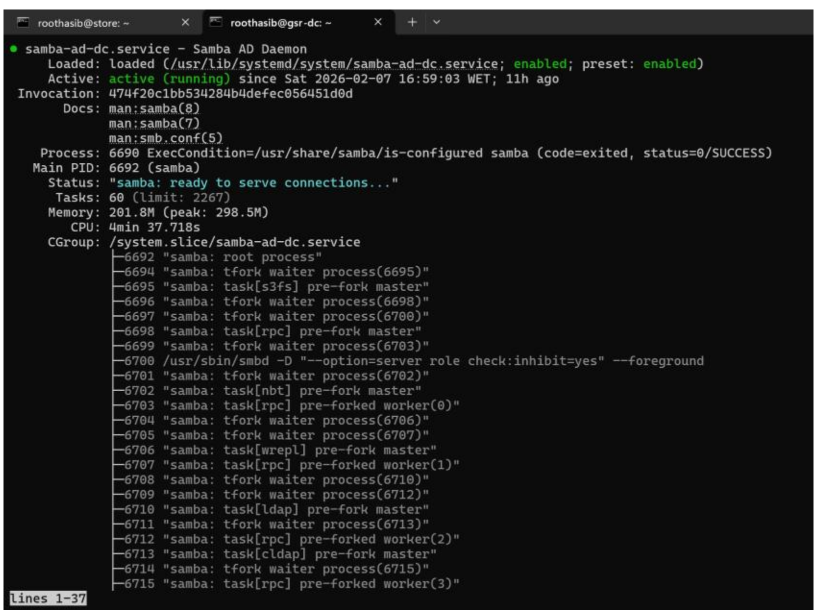
The domain controller was configured with hostname gsr-dc.gsr.ipb.local and static IP address 192.168.8.11/24.



[INSERT SCREENSHOT 1 (1b): Terminal showing hostname and IP configuration]

Samba Active Directory Domain Controller

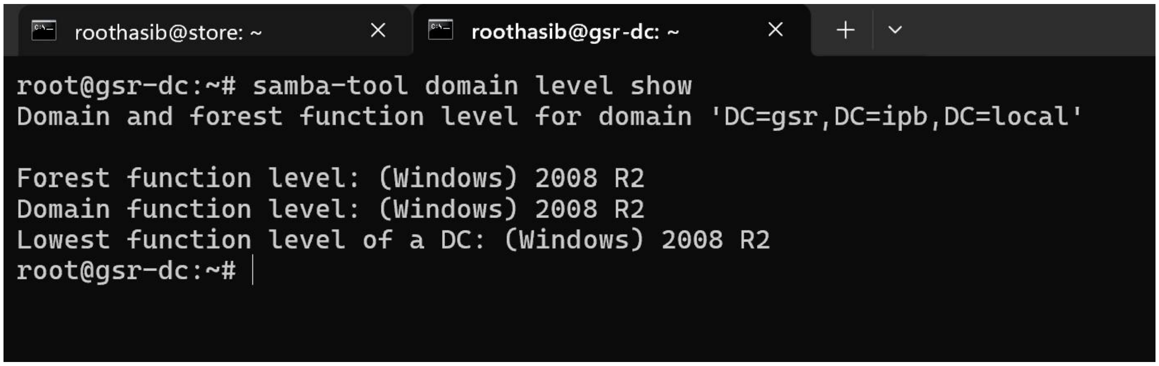
Samba AD DC was provisioned to create the gsr.ipb.local domain (NetBIOS name: GSR). The domain controller provides Active Directory services including user authentication, group policy, and centralized management for the organization.



[INSERT SCREENSHOT 2 (1b): Samba AD DC service status showing active state]

Domain Configuration

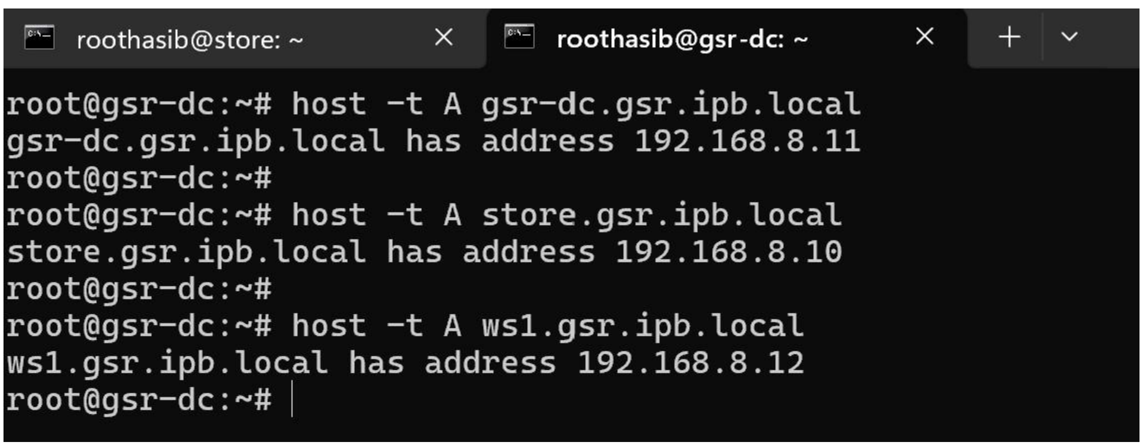
The domain was configured with appropriate functional levels for compatibility and security. The Samba internal DNS server was configured to handle name resolution for the domain.



[INSERT SCREENSHOT 3 (1b): Domain level information]

DNS Resource Records

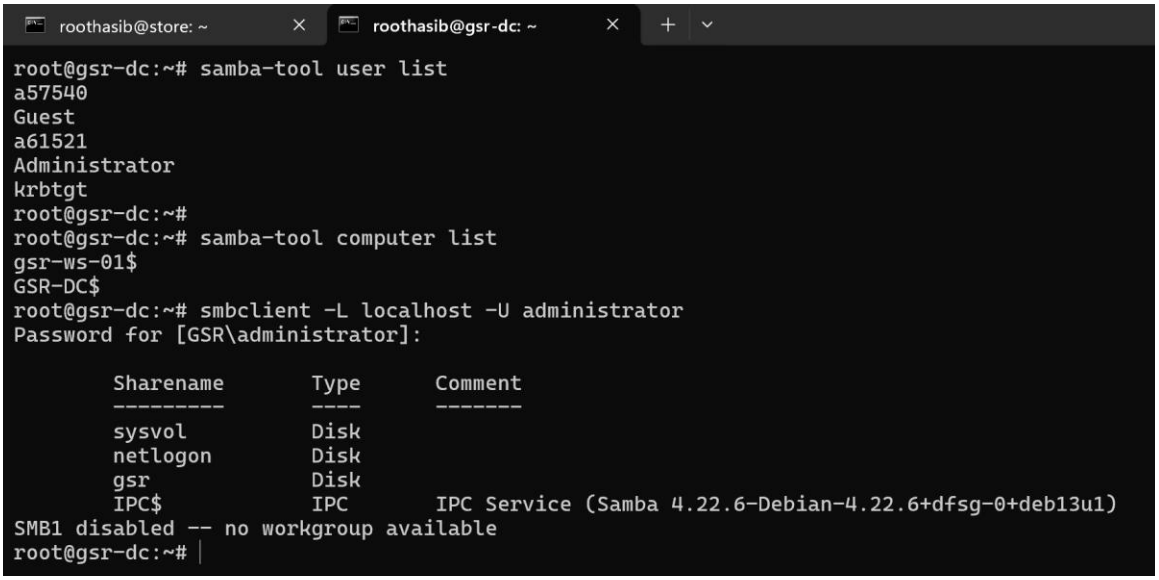
DNS A records were configured for all servers in the network infrastructure, enabling hostname-based communication between systems.



[INSERT SCREENSHOT 4 (1b): DNS query results for all three hosts]

User and Computer Account Management

Domain user accounts were created for both group members (a57540 and a61521) with appropriate permissions. Computer accounts were registered in Active Directory for domain-joined workstations.



```
roothasib@store: ~    roothasib@gsr-dc: ~
root@gsr-dc:~# samba-tool user list
a57540
Guest
a61521
Administrator
krbtgt
root@gsr-dc:~#
root@gsr-dc:~# samba-tool computer list
gsr-ws-01$
GSR-DC$
root@gsr-dc:~# smbclient -L localhost -U administrator
Password for [GSR\administrator]:

        Sharename       Type      Comment
        ---------       ----      -------
        sysvol          Disk
        netlogon        Disk
        gsr             Disk
        IPC$            IPC       IPC Service (Samba 4.22.6-Debian-4.22.6+dfsg-0+deb13u1)
SMB1 disabled -- no workgroup available
root@gsr-dc:~#
```

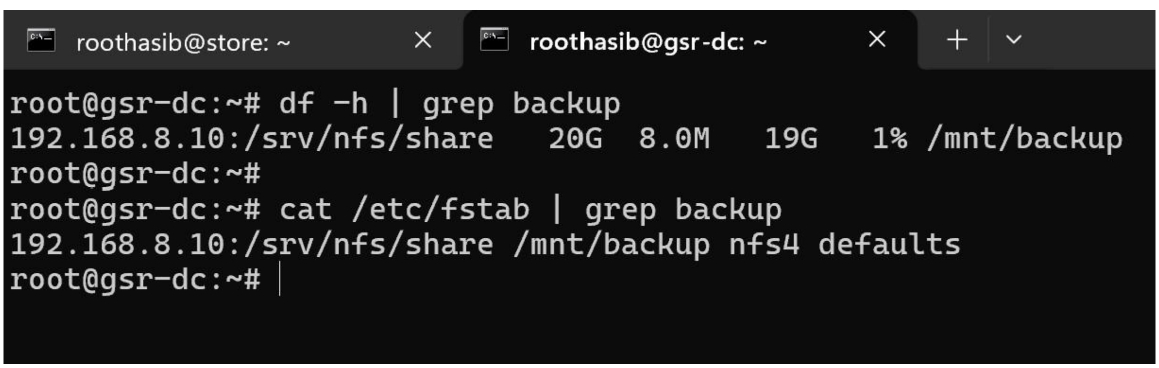
[INSERT SCREENSHOT 5 (1b): User list, computer list, and Samba shares - combined output]

Shared Folder Configuration

A shared folder \\gsr-dc.gsr.ipb.local\gsr was created and configured to be accessible by all domain users with read/write permissions. This shared resource allows domain users to store and access files from any domain-joined computer.

NFS Client and Backup Configuration

The domain controller was configured as an NFS client, mounting the shared storage from store.gsr.ipb.local at /mnt/backup. The mount was made persistent by adding an appropriate entry to /etc/fstab, ensuring the share is automatically mounted at system boot.

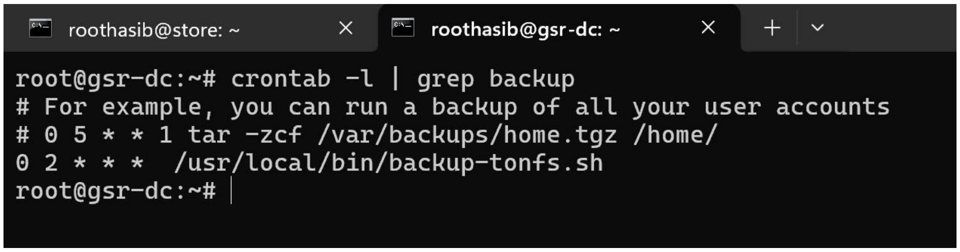


```
roothasib@store: ~    roothasib@gsr-dc: ~
root@gsr-dc:~# df -h | grep backup
192.168.8.10:/srv/nfs/share   20G  8.0M   19G   1% /mnt/backup
root@gsr-dc:~#
root@gsr-dc:~# cat /etc/fstab | grep backup
192.168.8.10:/srv/nfs/share /mnt/backup nfs4 defaults
root@gsr-dc:~#
```

[INSERT SCREENSHOT 6 (1b): NFS mount verification and fstab entry]

Automated Backup System

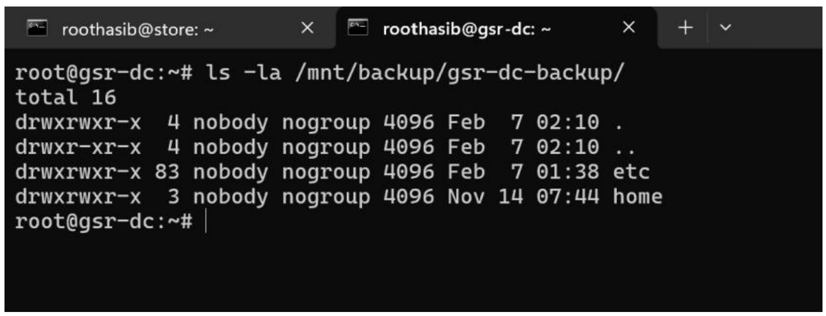
A daily backup job was configured using cron to execute at 2:00 AM every day. The backup script uses rsync to synchronize the /etc and /home directories to the NFS-mounted backup location, ensuring system configurations and user data are regularly backed up.



```
roothasib@store: ~        roothasib@gsr-dc: ~
root@gsr-dc:~# crontab -l | grep backup
# For example, you can run a backup of all your user accounts
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
0 2 * * *  /usr/local/bin/backup-tonfs.sh
root@gsr-dc:~#
```

[INSERT SCREENSHOT 7 (1b): Cron job configuration]

The backup system successfully creates and maintains copies of critical system directories on the remote NFS storage.



```
roothasib@store: ~        roothasib@gsr-dc: ~
root@gsr-dc:~# ls -la /mnt/backup/gsr-dc-backup/
total 16
drwxrwxr-x  4 nobody nogroup 4096 Feb  7 02:10 .
drwxr-xr-x  4 nobody nogroup 4096 Feb  7 02:10 ..
drwxrwxr-x 83 nobody nogroup 4096 Feb  7 01:38 etc
drwxrwxr-x  3 nobody nogroup 4096 Nov 14 07:44 home
root@gsr-dc:~#
```

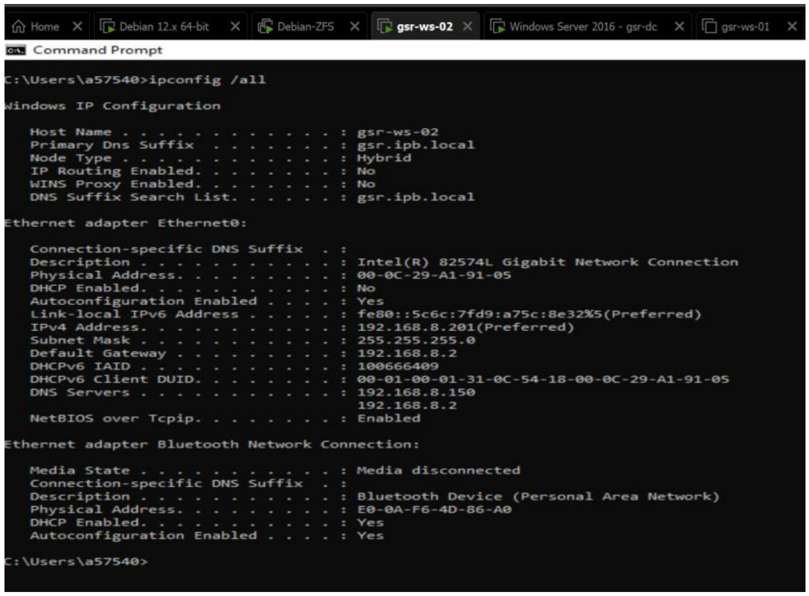
[INSERT SCREENSHOT 8 (1b): Backup directory contents showing etc and home folders]

1(c) Windows Workstation - gsr-ws-02

A Windows 10 Education workstation was configured and joined to the gsr.ipb.local domain, providing end-users with access to domain resources and centralized authentication.

Network Configuration

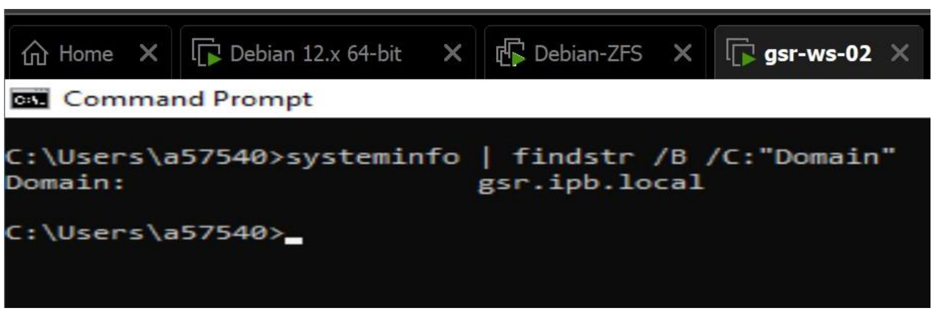
The workstation was configured with static IP address 192.168.8.201/24 on the local network with DNS pointing to the domain controller.



[INSERT SCREENSHOT 1 (1c): ipconfig /all output showing network configuration]

Domain Membership

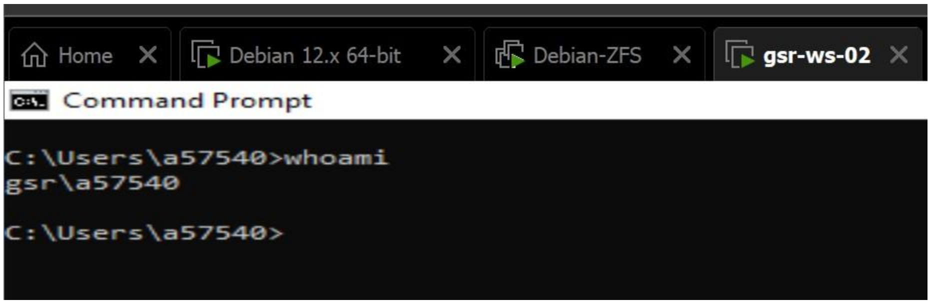
The workstation was successfully joined to the gsr.ipb.local domain, enabling centralized user authentication and access to domain resources.



[INSERT SCREENSHOT 2 (1c): systeminfo output showing domain membership]

Domain User Authentication

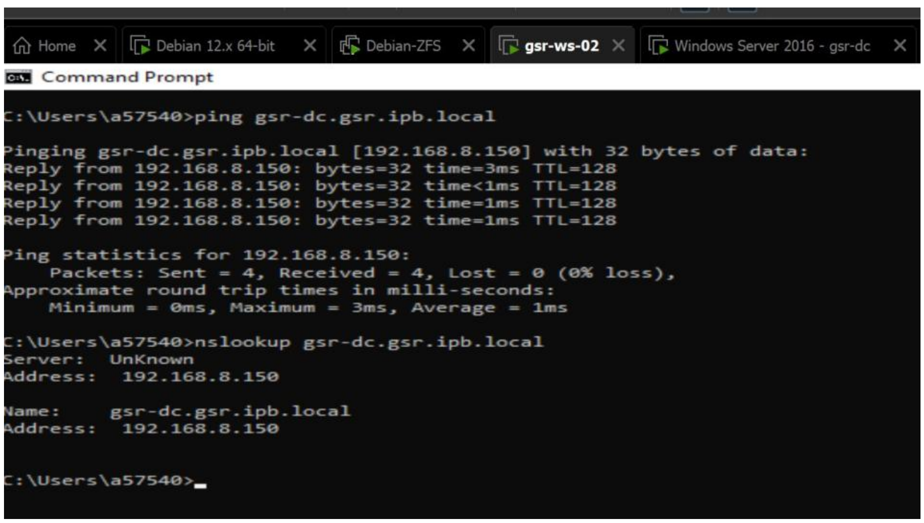
Domain users can successfully log into the workstation using their Active Directory credentials, demonstrating proper integration with the domain controller.



[INSERT SCREENSHOT 3 (1c): whoami output showing domain user]

Network Connectivity

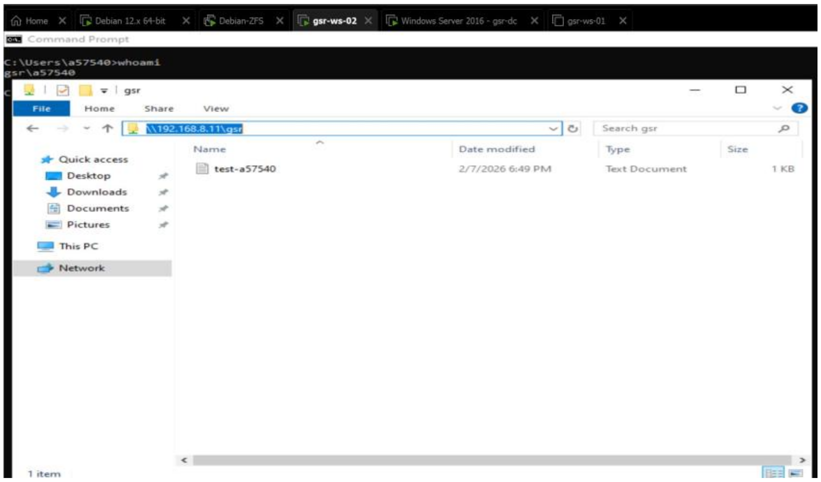
The workstation maintains network connectivity to the domain controller and other network resources.



[INSERT SCREENSHOT 4 (1c): ping and nslookup output]

Shared Folder Access

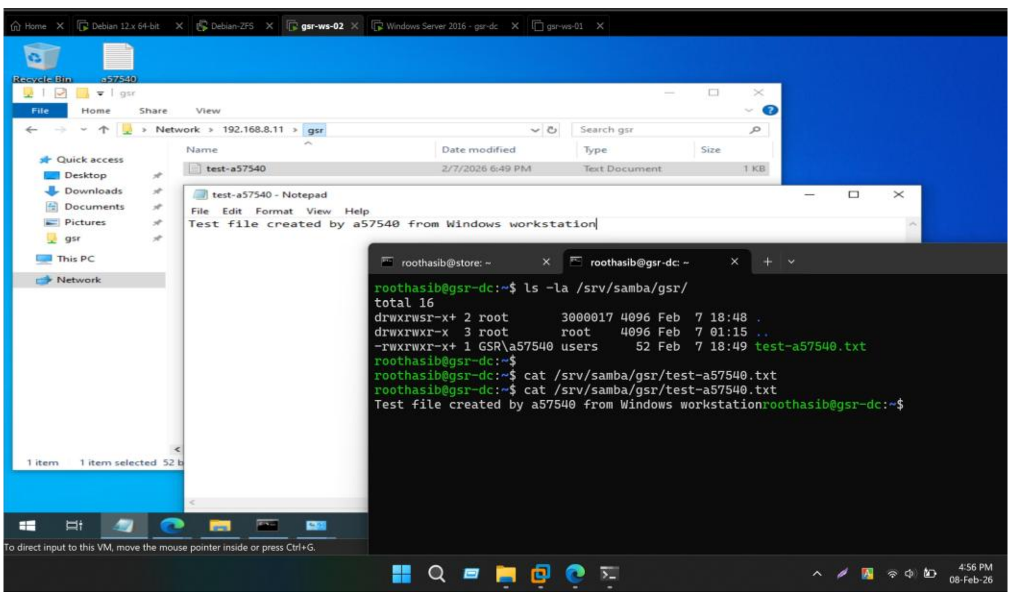
Domain users can successfully access the shared folder \\gsr-dc.gsr.ipb.local\gsr (accessible via IP \\192.168.8.11\gsr) and create files with appropriate permissions. Files created on the Windows workstation are properly stored on the Samba server with correct ownership attributes.



[INSERT SCREENSHOT 5 (1c): File Explorer showing access to shared folder]

File Creation Verification

To verify end-to-end functionality, a test file was created in the shared folder from the Windows workstation. This file was then verified on the domain controller, confirming proper Samba integration and file permissions.



[INSERT SCREENSHOT 6 (1c): Verification of test file on domain controller]

Question 2: Network Topology Explanation

2(a) Subnet, Netmask, and Gateway

The network infrastructure was implemented on a private IPv4 network with the following parameters:

- Subnet: 192.168.8.0/24
- Netmask: 255.255.255.0 (CIDR notation: /24)
- Gateway: 192.168.8.2

This subnet provides 254 usable IP addresses (192.168.8.1 through 192.168.8.254) for network devices. The /24 subnet mask was chosen as it provides sufficient address space for a small to medium enterprise environment while maintaining efficient network management. The gateway at 192.168.8.2 serves as the default route for external network access and internet connectivity.

2(b) IP Address Assignment and Configuration Method

The following table details the IP address assignments and configuration methods for all hosts in the network:

Hostname	IP Address	Configuration Method	Role
store.gsr.ipb.local	192.168.8.10	Static (Manual)	NFS Storage Server, SNMP, MRTG Monitoring
gsr-dc.gsr.ipb.local	192.168.8.11	Static (Manual)	Samba AD Domain Controller, DNS Server
gsr-ws-02 (ws1 role)	192.168.8.201	Static (Manual)	Windows 10 Workstation
Gateway/Router	192.168.8.2	N/A	Network Gateway

Configuration Method Rationale

All servers and the workstation were configured with static IP addresses rather than DHCP to ensure consistent network addressing for infrastructure services. Static addressing is essential for:

1. DNS stability: The domain controller must maintain a consistent IP address for DNS resolution
2. Service reliability: NFS mounts and Samba shares require predictable addressing
3. Administrative clarity: Static addressing simplifies troubleshooting and network documentation

The static IP configuration was implemented by editing /etc/network/interfaces on Debian systems and using the Network and Sharing Center on Windows 10.

Additional Software, Distributions, and Devices

The following software components and technologies were utilized to implement the requested infrastructure:

Operating Systems:

- Debian 13 (Trixie) - Used for both store.gsr.ipb.local and gsr-dc.gsr.ipb.local servers
- Windows 10 Education - Used for the workstation (gsr-ws-02)

Server Software and Services:

- Samba 4.22.6 - Active Directory Domain Controller implementation
- NFS kernel server (nfs-kernel-server) - Network File System server for shared storage
- Net-SNMP - SNMP monitoring agent with v3 authentication
- MRTG (Multi Router Traffic Grapher) - Network and system monitoring with graphical output
- Apache2 Web Server - Hosting the MRTG web interface
- rsync - File synchronization tool for automated backups
- BIND9 DNS - Integrated with Samba for domain name resolution

Network Protocols:

- NFSv4 - Network File System protocol for shared storage
- SNMPv3 - Secure network monitoring with authentication and encryption
- SMB2/SMB3 - Modern file sharing protocol (SMB1 disabled for security)
- LDAP - Directory services protocol for Active Directory
- Kerberos - Authentication protocol for domain security
- DNS - Domain Name System for hostname resolution

Virtualization Platform:

- VMware Workstation - Virtualization platform hosting all three virtual machines

Network Configuration:

- NAT Networking - VMware NAT network providing connectivity to all VMs

The NAT network simulates a physical network switch/router environment with internet access through the host system

Administrative Tools:

- samba-tool - Command-line tool for Samba AD DC management
- systemctl - Service management on Debian systems
- cron - Task scheduler for automated backups

This infrastructure demonstrates a complete small business IT environment with centralized authentication, shared storage, monitoring, and automated backup capabilities suitable for organizations with 10-50 users.

Conclusion

This assignment successfully implemented a complete SME infrastructure including:

- Centralized storage with NFS and SNMP monitoring
- Active Directory domain controller with DNS services
- Domain-joined Windows workstation with shared folder access
- Automated backup system for critical data
- Network monitoring and reporting capabilities

All functional requirements were met, and the system demonstrates enterprise-grade features including secure authentication, centralized management, and data redundancy suitable for small to medium business environments.